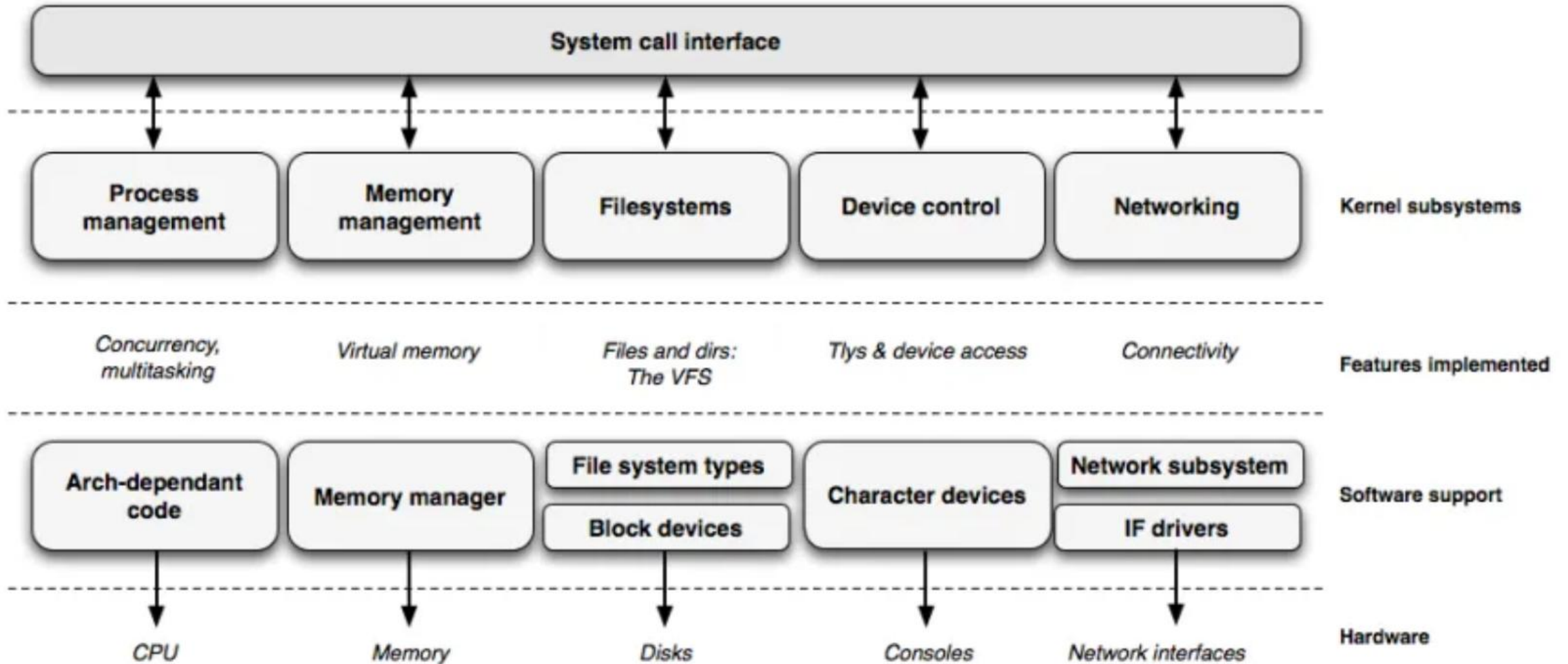


Linux Kernel Architecture

Linux Kernel Architecture Overview



- The Linux Kernel architecture is composed of several layers, each with a specific purpose. The main layers are:
- **User Space:** This is where user applications and libraries reside. User Space communicates with the Kernel through system calls, making it easy for applications to request services from the Kernel without direct hardware interaction.

- **System Call Interface:** This layer acts as a gateway between User Space and Kernel Space. It processes system calls from user applications and invokes the appropriate Kernel functions (will be covered under file operations).

- **Kernel Space:** This is where the core Kernel components reside. Kernel Space is further divided into several subsystems, such as Process Management, Memory Management, File System Management, Device Management, and more.

User Space vs Kernel Space

Sr No	User Space	Kernel Space
1	The memory area where user applications run.	The memory area where the Linux kernel runs and provides its core services.
2	Restricted access — can't directly interact with hardware or kernel memory.	Full access to hardware and system resources.

User Space vs Kernel Space

Sr No	User Space	Kernel Space
3	Memory crashes affect only the process. Faults in user space don't crash the whole OS	Memory crashes can hang the system